Prototype 1 Rectifier Validation

Objective: Verify three phase rectifier operation by measuring and graphing generator input voltage (V_{L-L}), generator frequency ($f_{gen)}$, and open circuit DC bus voltage (V_{DCOC}).

Scope: PMSG, Three-Phase Rectifier, DC Bus (open circuit)

Equipment: Oscilloscope, DC motor, PMSG, PCBA

Setup:

1. Configure DC motor to spin the PMSG
2. Connect PMSG phases to screw terminal J1 in ABC order from left to right
3. Configure an oscilloscope probe to measure V_{L-L}.
4. Configure an oscilloscope probe to measure V_{DCOC} using screw terminal J4 for the DC bus and GND connection (left and right, respectively)

Test Procedure:

Use the DC motor to spin the PMSG at increasing intervals of 10Hz until reaching 100Hz or V_{L-L} = 155V. At each interval, use an excel sheet to document f_{gen}, V_{L-L}, and V_{DCOC}. The frequency should be determined using the V_{L-L} oscilloscope waveform.